

Not All Bad Demonstrations Are Equally Bad: Quantifying How Demonstration Failure Modes Degrade Closed-Loop Policy Performance

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<https://github.com/imjbassi/demo-quality-robustness>

September 2026

Abstract

Demonstration quality is often treated as a preprocessing concern in imitation learning. We instead hold architecture, hyperparameters, episode count, and evaluation conditions fixed while varying the type and prevalence of defective demonstrations. Five failure modes—occlusion, accidental success, corrective flailing, truncation, and inconsistent strategy—are introduced into behaviour-cloning datasets for a controlled 2D block-pushing task. Across 78 policy fits and three training seeds, outcomes differ sharply by defect type. At full contamination, mean closed-loop success is 0.963 for corrective flailing but 0.012 for accidental success, compared with 0.972 for clean data. Accidental-success data retains mean success of 0.890 at 75% contamination before collapsing at 100%. Pooled open-loop action-prediction error correlates strongly with closed-loop success ($r = -0.91$), but within-mode correlations range from -0.15 to -0.94 ; in the descriptive band $[0.20, 0.30]$, similar open-loop errors coincide with closed-loop success from 0.315 to 0.980. A control analysis also rejects our initial interpretation of the inconsistent-strategy result: the apparent degradation is explained by the alternate strategy’s lower clonability, not by mixing itself. These simulation results motivate defect-specific quality control and closed-loop validation, while the small seed count and simplified dynamics limit quantitative generalisation. Code, raw results, and analysis are publicly available.

1 Introduction

Behaviour cloning is a standard baseline for robot manipulation [1], and its central known weakness is compounding error: small action errors move the policy off the demonstration distribution, where its predictions degrade further [2]. The standard responses to this are architectural or algorithmic: better policy classes [5], or interactive data collection.

Practitioners running data-collection pipelines know something the literature under-quantifies: the demonstrations themselves fail in structured, recognisable ways. A camera is occluded mid-episode. An operator fumbles and the object lands on the goal anyway. Two operators solve the same task with incompatible strategies. An episode is cut short. These are not Gaussian noise on a clean signal; they are distinct defects with distinct structure, and quality-control effort spent on one

is effort not spent on another.

Prior work has established that demonstration quality matters in aggregate [3, 4]. What is less established is a *ranking*: given finite QC budget, which defect should a pipeline prioritise? And, because closed-loop evaluation is expensive and open-loop action-prediction error is cheap, does the cheap metric even detect the damage?

This paper answers both questions in a deliberately small, fully controlled setting. Our contributions are:

1. A controlled experimental design treating demonstration quality as the independent variable, with contamination rate as a single comparable axis across five distinct failure modes.
2. A descriptive ranking of failure modes by closed-loop degradation in a controlled block-pushing setting.
3. Evidence that a strong pooled association between open- and closed-loop metrics does not imply uniform diagnostic value within each failure mode.
4. A control analysis showing that an apparent strategy-inconsistency effect is instead explained by unequal strategy clonability.

2 Method

2.1 Environment

Push2D is a 2D block-pushing task implemented in vectorised NumPy. A point gripper must move a block into a goal region. Pushing was chosen over reaching deliberately: it imposes a *staging* requirement, since the gripper must position itself behind the block relative to the goal before pushing is productive. This makes compounding error consequential. A policy that drifts during approach ends up on the wrong side and pushes the block *away* from the goal, which is unrecoverable within the episode budget. This is precisely the class of failure that open-loop metrics are structurally poor at surfacing, since each individual action error is small.

Observations are 10-dimensional (gripper, block and goal positions plus two relative vectors); actions are 2D displacements clipped to $[-1, 1]$. Episodes run for at most 110 steps and terminate on success, defined as the block centre falling within 0.05 of the goal.

2.2 Scripted experts

Using scripted rather than human demonstrations is what makes the experiment controlled: corruption can be applied with exact, programmatic ground truth rather than human judgement about what counts as a bad demo.

The **primary expert** is a hand-coded orbit-then-push controller that circles the block at a safe radius until aligned, then pushes toward the goal. Measured success rate: **100.0%**.

The **alternate expert** always circles in the same rotational direction from a wider staging radius, producing a visibly different route through state space. It was explicitly tuned until it *also* reaches **100.0%**, so that mixing the two tests strategy *inconsistency* rather than strategy *quality*. This tuning turns out to matter substantially (Section 4).

The **naive controller** charges blindly at the block and never reasons about the goal; the block travels along whatever ray the gripper started on. It succeeds on approximately 6% of episodes, purely by geometric luck.

2.3 Failure modes

Each mode produces demonstrations that a real pipeline could plausibly log:

Occlusion. A contiguous window covering 35% of timesteps has its observations frozen at the last visible frame. Actions remain expert-quality; the demonstrator can see, the camera cannot. The resulting transitions have corrupted inputs and pristine labels.

Accidental success. Rollouts from the naive controller, keeping only those that reached the goal. Correct outcome, no correct behaviour anywhere in the episode.

Corrective flailing. Heavy uniform jitter on the first 12 timesteps, after which the demonstrator settles into correct motion and still succeeds.

Truncation. A clean episode cut at 70% of its length and logged anyway, what a QC filter missing a premature stop looks like.

Inconsistent strategy. Complete episodes from the alternate expert. Every one is a genuine success; the only defect is disagreement with the primary strategy about what to do in a given state.

2.4 Two design decisions

Action-level corruptions are re-simulated, not pasted. Overwriting actions in a logged trajectory without re-running the dynamics yields observations that no longer follow from the actions, a physically impossible episode. That measures label noise, not bad demonstrations. Every mode that touches actions replays the episode through the environment.

Corrupted episodes must pass the same success-based QC filter a real pipeline applies. Modes are rejection-sampled to successful episodes only. Truncation is the deliberate exception, since it *is* the filter failing.

2.5 Sweep axis, policy and evaluation

The sweep axis is **contamination rate** ρ , not per-episode severity. Every dataset contains the same number of episodes; ρ of them are defective and $(1 - \rho)$ are clean, with per-episode severity held fixed. This places all five modes on one comparable axis and answers the question a pipeline actually asks: what fraction of the dataset can be this kind of bad before the policy breaks?

We sweep $\rho \in \{0, 0.25, 0.5, 0.75, 0.9, 1.0\}$ across three training seeds. The shared clean baseline is fitted once per seed, giving 78 distinct policy fits: three clean fits and $5 \times 5 \times 3$ contaminated fits. Every run uses an identical 2×256 ReLU MLP trained with MSE behaviour cloning for 50 epochs on 400 episodes ($\approx 13k$ transitions). The model is deliberately small: it is not the independent variable.

Two metrics are recorded per policy. **Open-loop MSE** is action-prediction error against the expert on held-out *clean* expert states (cheap, offline, no environment interaction). **Closed-loop success rate** runs the policy in the environment over 200 episodes. The same 200 evaluation initial conditions are reused for every configuration to support paired descriptive comparisons.

3 Results

The clean baseline reaches 0.972 ± 0.010 closed-loop success (mean $\pm$ standard deviation over three training seeds). All uncertainty summaries are descriptive; the small seed count does not support formal significance claims.

3.1 Failure modes are not interchangeable

Table 1 gives the headline comparison. At full contamination, corrective flailing reduces mean success by 0.009 relative to the clean baseline, whereas accidental success reduces it by 0.960. Both are “bad demonstrations” under a practical taxonomy, yet their observed costs differ by two orders of magnitude.

Corrective flailing causes little degradation in this setting. Jitter-then-recover trajectories may provide DAgger-like coverage of off-distribution states [2]; at $\rho = 1.0$, mean success is 0.963, close to the clean baseline. This pattern makes corrective flailing a lower QC priority here, although three seeds are insufficient to establish equivalence or benefit.

Truncation does not reduce mean success in the observed setting: mean success is 0.987 at $\rho = 1.0$. A plausible explanation is that the push action can be extrapolated from the retained approach phase. This interpretation should not be transferred to tasks with a distinct terminal phase, such as grasping, without further experiments.

3.2 Accidental success is a cliff, not a slope

The most operationally significant observed result is the shape of the accidental success curve: 0.965 at $\rho = 0.5$, 0.890 at $\rho = 0.75$, 0.747 at $\rho = 0.9$, and **0.012** at $\rho = 1.0$.

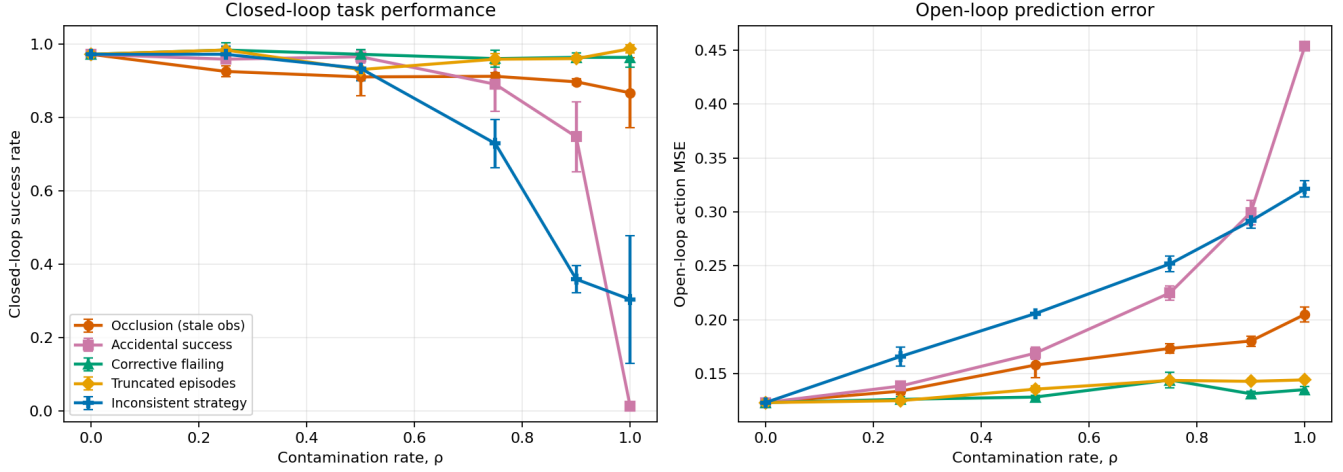


Figure 1: Dose-response curves across contamination rate. **Left:** closed-loop success rate, the quantity of interest. **Right:** open-loop MSE, the offline proxy. Accidental-success policies retain high mean success at $\rho = 0.75$ before collapsing at $\rho = 1.0$, whereas corrective flailing and truncation remain comparatively stable. Points are means and error bars are standard deviations over three training seeds.

Failure mode	$\rho=0.5$	$\rho=1.0$	Pearson r
Corrective flailing	0.972	0.963	-0.35
Truncated episodes	0.930	0.987	-0.15
Occlusion (stale obs)	0.910	0.867	-0.58
Inconsistent strategy	0.933	0.303	-0.90
Accidental success	0.965	0.012	-0.94
Clean baseline	0.972	n/a	

Table 1: Closed-loop success rate by failure mode and contamination rate, mean over three seeds. Pearson r is the within-mode correlation between open-loop MSE and closed-loop success across all runs of that mode.

In this experiment, a minority of clean demonstrations masks much of the problem. A pipeline monitoring only success-labelled demonstration throughput would not reveal the approaching policy failure. The diagnostic wrong-side push rate tells the same story, rising from 0.008 at $\rho = 0.5$ to 0.468 at $\rho = 1.0$: the policy has learned to shove the block in essentially arbitrary directions.

3.3 Open-loop evaluation has failure-mode-dependent value

Pooled across all 78 distinct policy fits, open-loop MSE correlates with closed-loop success at $r = -0.91$. This strong aggregate association could appear to justify its use as a screening metric.

Broken out by failure mode, however, Pearson correlations range from -0.15 (truncation) to -0.94 (accidental success). The metric is strongly associated with success for some defects but weakly associated for truncation and corrective flailing, precisely where closed-loop performance varies little.

A complementary descriptive comparison considers the 15 runs whose open-loop MSE falls in the descriptive band

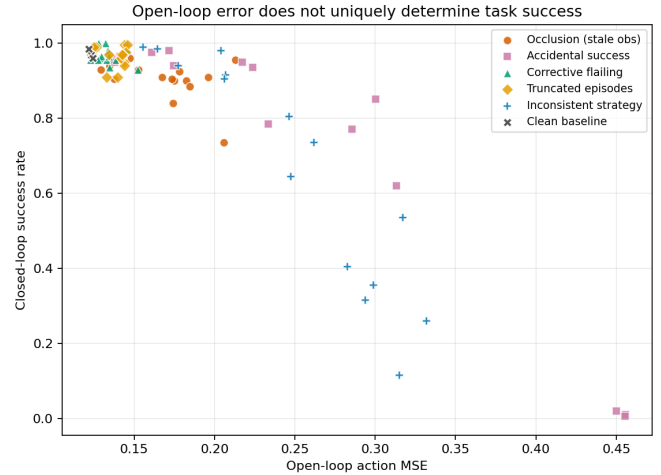


Figure 2: Open-loop MSE against closed-loop success for all 78 distinct policy fits. Within the descriptive MSE band $[0.20, 0.30]$, closed-loop success spans 0.665.

$[0.20, 0.30]$. Their closed-loop success ranges from **0.315** to **0.980**. This band is not a statistical equivalence test; it illustrates that numerically similar offline scores need not determine similar task outcomes. Figure 2 shows the full distribution, including the high-MSE collapse points that contribute strongly to the pooled correlation.

4 A prediction that failed

We predicted that mixing two valid strategies would cause multimodal-averaging collapse, since an MSE-regression policy averages conflicting action targets by construction. The data does not support this, and the control is worth stating explicitly because it nearly produced a false positive.

At $\rho = 1.0$ the dataset is entirely alternate-strategy and

therefore fully self-consistent; there is no mixing left to blame. Success there (0.303) is the *clonability floor* of that strategy, not evidence of inconsistency harm. Both experts solve the task on 100% of episodes; the alternate is simply harder to clone, having a longer horizon and more orbit steps over which error compounds.

Reading intermediate ρ against the line joining the two endpoints isolates any harm attributable to mixing:

ρ	Observed	Interpolated	Residual
0.25	0.972	0.805	+0.167
0.50	0.933	0.637	+0.296
0.75	0.728	0.470	+0.258
0.90	0.358	0.370	-0.012

Table 2: Observed success relative to linear interpolation between the $\rho = 0$ and $\rho = 1$ endpoints. Positive residuals indicate performance *above*, not below, the clonability-only interpolation.

The residual is non-negative at three of four intermediate points and slightly negative at $\rho = 0.9$. Thus, the observations provide no consistent evidence that mixing is worse than the clonability-only interpolation; at lower contamination rates, clean data may instead be protective. The inconsistent-strategy curve in Figure 1 looks damaging only because one of the two strategies is intrinsically harder to imitate.

This control materially qualifies the headline comparison. Removing the inconsistent-strategy runs from the matched-MSE band narrows the closed-loop spread from 0.665 to approximately 0.22. This narrower descriptive spread is considerably more modest than the raw figure suggests.

5 Limitations

Three training seeds are insufficient for precise inference. Some cells are noisy: occlusion at $\rho = 1.0$ spans 0.735 to 0.955 across seeds. The means and standard deviations therefore describe these runs rather than establish population-level effects; confirmatory work should use more seeds and report uncertainty intervals.

One fixed evaluation set. Reusing the same 200 initial conditions reduces between-configuration nuisance variation but does not measure sensitivity to the evaluation-state distribution. Replication should vary both training and evaluation seeds.

A 2D toy environment. Whether these rankings survive in a 7-DOF setting with contact dynamics and visual observations is open. This work establishes a methodology, not transferable constants.

One policy class. MSE regression averages multimodal action distributions by construction. The inconsistent-strategy result in particular may be specific to this choice; replicating it with a diffusion policy [5] is the most informative single follow-up.

Transition imbalance. Alternate-strategy episodes are longer, so at fixed episode count they contribute more transitions. A transition-matched replication would tighten that arm further.

6 Conclusion

Treating demonstration quality as the independent variable, rather than an implementation detail, produces a defect-specific ranking in this controlled task: accidental-success performance collapses near full contamination, occlusion produces a moderate decline, and corrective flailing and truncation remain comparatively stable. The practical hypothesis for larger systems is that QC effort should be allocated by defect type rather than uniformly.

The evaluation result is equally important: open-loop action-prediction error has a strong pooled association with closed-loop success, yet its diagnostic value varies substantially by failure mode. Offline screening is useful, but it is not a substitute for closed-loop validation.

Reproducibility statement

Source code, exact training and evaluation seeds, raw per-run results, and the scripts used to generate both figures are available at <https://github.com/imjbassi/demo-quality-robustness>. The repository also records the software requirements and commands needed to reproduce all 78 policy fits and rebuild this manuscript.

References

- [1] D. Pomerleau. ALVINN: An Autonomous Land Vehicle in a Neural Network. *NeurIPS*, 1989.
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